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POWER SUPPLY FOR A CLOCK
Using Supercapacitor And Small Solar Panels

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Solar photovoltaic panels produce energy only while the Sun shines, but lighting is required mainly during night. So, the solar energy needs to be stored, which is now generally done through lithium-ion batteries.

The lithium-ion batteries have some limitations. For one, cost of lithium-ion batteries is increasing year after year, so these are suitable for high-priced mobile items like vehicles and smartphones only. For stationary applications we need to find alternatives.

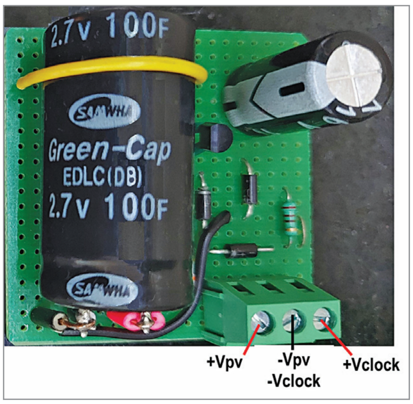

Fig. 1: Author's prototype of clock power supply using solar PV and supercapacitor

Second, lithium-ion batteries can handle a few thousand charge/discharge cycles only. This means the batteries need to be replaced after about five years of use.

Third, re-cycling of lithium-ion batteries is challenging and expensive. Also, India does not have lithium mines, so we are completely dependent on imports.

Fourth, lithium-ion batteries also use metals like nickel and cobalt. Therefore, these batteries are not environment-friendly. Besides, lot of mining is required to get these metals.

Hence, it becomes important to explore other devices for energy storage. Here we will explore use of supercapacitors for energy storage for low power but widespread applications.

Supercapacitors (or ultracapacitors) are also called electric double-layer capacitors. They use porous carbon electrodes and store energy in the form of electric field. Unlike batteries, which store energy in electrochemical form, these offer high-value capacitance, which can be measured in Farads. Typical working voltage of these capacitors is low (2.7V).

As of today, the cost of supercapacitors is high. Just for storing a few watt-hours of energy we have to use several capacitors. Hence, the cost of even small low-power systems is fairly high.

However, there are some ways the cost of supercapacitors can be reduced, a couple of which are suggested below.

Mass production. As discussed earlier, supercapacitors do not need expensive raw material. So, their high cost is mainly due to their low volumes of production. Therefore, there is a need to identify applications that can generate big demand for them. As their demand and production up, automatically the manufacturing cost will come down.

Redesign for slow charge/discharge applications. Most supercapacitors available in the market are designed for handling high (in amps) charge/discharge currents. Therefore, these supercapacitors have thick terminals and large electrodes. However, in small solar applications the capacitor charge/discharge currents are low (in milliamps). Hence, for such applications capacitor design needs to be optimised to reduce cost and size.

Supercapacitor based clock is one of the best suited applications. Clocks use low-power AA primary batteries, which need to be replaced almost every year. Every house has a few of such clocks. So, we consume a large number of

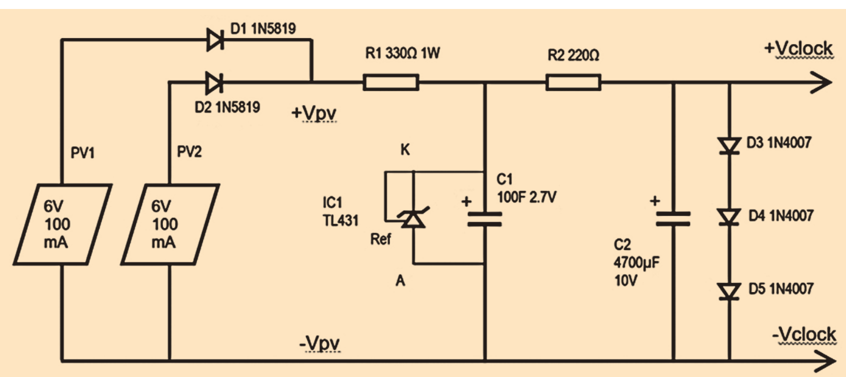

Fig. 2: Circuit diagram of the clock power supply